MIC-501T SUBJECTIVES

MID TERM- 100%

MICROBIOLOGY

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SUBJECTIVES

Branches of mycology:

Mycology is the branch of biology concerned with the study of fungi, including their genetic and biochemical properties, their taxonomy and their use to humans as a source for tinder, traditional medicine, food, and entheogens, as well as their dangers, such as toxicity or infection. Branches: 1= Medical mycology, 2= Agricultural mycology, 3= Ecological mycology – Decomposers of woods, leaves, feces and other organic matters.

Define Dry heat method of sterilization:

Dry heat sterilization methods include hot-air ovens, flaming (using a Bunsen burner for items like platinum loops), and incineration (burning to ashes). These techniques eliminate microorganisms by exposing them to elevated temperatures without moisture, ensuring the destruction of pathogens through oxidative processes or direct burning.

Define Five Kingdom Classification of Organisms.

The Five Kingdom Classification classifies organisms into five groups: Monera (prokaryotic bacteria), Protista (single-celled eukaryotes), Fungi (multicellular, absorptive heterotrophs), Plantae (multicellular, photosynthetic autotrophs), and Animalia (multicellular, ingestive heterotrophs). This system, proposed by R.H. Whittaker, categorizes living organisms based on their cellular structure, nutrition, and reproduction methods.

Define Spontaneous Generation:

(Aristotle) Spontaneous generation was a once-held belief that living organisms could arise spontaneously from non-living matter. This concept, prevalent until the 19th century, proposed that life could emerge without pre-existing life. However, advancements in microbiology, particularly Pasteur's experiments, disproved this notion, establishing the principle of biogenesis.

Who name the small pox and when?

Lady Mary Wortley Montagu named the smallpox in 1717, while Edward Jenner developed the smallpox vaccine in 1796.

Give Five Types of Microscopy:

1= Darkfield Microscopy: Enhances contrast by illuminating specimens against a dark background., 2= Phase Contrast Microscopy: Accentuates phase shifts in light passing through transparent specimens., 3= Fluorescent Microscopy: Uses fluorescent dyes to visualize specific cellular structures., 4= Transmission Electron Microscopy: Achieves high resolution by transmitting electrons through thin specimens., 5= Scanning Electron Microscopy: Provides detailed surface images using electrons.

Enlist steps of Gram Staining:

1. Primary Stain (Crystal Violet): Colors all cells.
2. Mordant (Iodine): Enhances crystal violet retention.
3. Decolorization (Alcohol or Acetone): Removes stain from some cells.
4. Counterstain (Safranin): Colors decolorized cells.
5. Wash and Dry: Final steps to observe Gram-positive (purple) and Gram-negative (pink) bacteria.

Define Metachromatic granules:

Metachromatic granules, also known as volutin, are large inclusions within bacterial cells. They stain red with methylene blue due to their unique property of displaying color changes during staining. Composed of inorganic phosphates, these granules are characteristic features observed in bacteria, such as Corynebacterium diphtheriae.

Which factors can influence the Bacterial death rate?

Bacterial death rate can be influenced by factors such as temperature, pH levels, exposure to antimicrobial agents, the concentration of nutrients, and the presence of reactive oxygen species. These conditions affect bacterial metabolism and susceptibility to environmental stressors.

Enlist chemicals which disrupts the plasma membrane.

1= Hexachlorophene, 2= triclosan, 3= Chlorhexidine, 4= Alexidine.

Define Gene, chromosome, genotype, phenotype.

1= Gene: The basic unit of heredity passed from parent to child. 2= Chromosome: threadlike structures made up of protein and DNA that carry genomic information from cell to cell. 3= Genotype: Genotype is the genetic makeup of an individual cell or organism that determines or contributes to its phenotype, 4= Phenotype: all observable characters that influenced by genotypes and environment.

Define any Three shapes of bacteria?

1= Coccus These are circular or spherical in shape. 2= Bacillus These rod-shaped bacteria. 3= Spiral These are curved shaped bacteria. They are further divided into 3 more subgroups. Vibrio: curved rods Spirillum: Helical but rigid Spirochete: Helical but flexible Spirochetes move by axial filaments which are enclosed by the outer membrane.

Write the name of five layers of endospore.

1) Exosporium: A thin delicate outermost covering of the spore 2) Spore coat: 2nd layer underneath the Exosporium. It is thick and composed of several protein layers 3) Cortex: It is the 3rd layer from outside in. It has peptidoglycan in it. 4) Spore cell wall or core wall: Surrounds the protoplast or spore core 5) Spore core: Contains nucleoid and ribosomes.

Difference between death phase and stationary phase of bacterial growth.

1= The Stationary Phase: This is also called a period of equilibrium as microbial deaths equal production of new cells. In other words, organisms start dying during this phase, however, the number of dead organisms is replaced by new organisms because there is still replication of cells going on. So, overall number of organisms does not change. This is the reason; it is called a stationary phase.

2= The Death Phase: The number of deaths exceeds the number of new cells formed during this phase. In other words, overall number decreases. It is also called logarithmic decline phase. Why there is a decline phase or death phase. The reason is simple: Nutrients are depleted and waste products which are toxic to the cells accumulate suppressing the growth and killing the cells.

Define Pasteurization:

Pasteurization is a heat treatment process that aims to reduce microbial contamination in food and beverages. Named after Louis Pasteur, it involves heating liquids (like milk or juice) to a specific temperature (commonly around 72°C for 15 seconds) and then rapidly cooling to kill or inactivate harmful microorganisms, notably pathogenic bacteria. This process preserves the product's quality while extending shelf life and enhancing safety. Unlike sterilization, pasteurization does not eliminate all microorganisms but significantly reduces their numbers to prevent spoilage and mitigate health risks. The following three equivalent heat treatments can be given to achieve pasteurization of milk: 1= 63°C for 30 min, 2= High-temperature short-time: 72°C for 15 sec, 3= Ultra-high-temperature: 140°C for 4 sec.

Define Ethylene Oxide:

Many heat-sensitive items such as disposable plastic, petri dishes and syringes, heartlung machine components, sutures, and catheters are sterilized by ethylene oxide. Ethylene oxide is a strong alkylating agent that kills by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It rapidly penetrates packing materials, even plastic wraps. EtO is explosive, supplied in a 10% to 20% concentration mixed with either CO2 or dichlorodifluoromethane. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/L.

How Osmotic pressure causes plasmolysis.

Osmotic pressure causes plasmolysis by drawing water out of the cell through the process of osmosis. When a cell is placed in a hypertonic solution, water molecules move out of the cell, leading to shrinkage and detachment of the plasma membrane from the cell wall.

Define plasmolysis osmotic pressure?

1= Plasmolysis: Plasmolysis is the process in which cells lose water in a hypertonic solution. 2= Osmotic Pressure: Pressure required to prevent movement of water into a solution containing some solute.

Define the terms Antisepsis, sepsis, sterilization. 1= Antisepsis: Removing pathogens from living tissue. 2= Sepsis: Microbial contamination A toxic inflammatory condition arising from the spread of an organism from a focus of infection. 3= Sterilization is the complete elimination of all microbial life, including spores, ensuring aseptic conditions. Sanitization: Lowering microbial counts on eating utensils.

What are physical requirements for the growth of bacteria?

Physical requirements for bacterial growth include temperature, pH, osmotic pressure, and oxygen concentration. Bacteria thrive within specific ranges of these factors, influencing their metabolic processes and reproduction.

What are the Uses of Growth Curve:

The growth curve is employed to study the growth dynamics of bacteria, ensuring optimal conditions for mass production during the exponential phase. It is crucial for various applications, including vaccine production, obtaining bacterial seed cultures, and preserving microbial cultures for scientific research and industrial processes.

Define coenzymes with examples:

Coenzymes are molecules that assist enzymes in catalyzing reactions. Functioning as electron carriers, they receive electrons from nutrients and donate them in subsequent reactions.

Examples include NAD+, NADP+, FMN, and FAD, essential in redox reactions and energy transfer processes within cells.

Define Stages of Germination of Spore:

The stages of spore germination involve 1= activation, where the spore becomes metabolically active; germination, with the emergence of a germ tube; and outgrowth, where the germinated spore develops into a vegetative cell or structure capable of growth and division.

Define the modes of killing bacteria:

Modes of killing bacteria include moist heat and dry heat, which alter the molecular structure of proteins and carbohydrates. Low temperatures inhibit bacterial growth. High pressure causes changes in molecular structure. Desiccation disrupts bacterial metabolism. These methods are employed to eliminate or control bacterial populations in various settings.

Define culture, culture medium, inoculum.

1= What is culture. This term is used to define growth of microbes. In other words, microbial growth in the lab is called a culture. 2= What is a culture medium? A nutrient material that supports the growth of microbes in the lab is called a culture medium. A culture media is a special medium used in microbiological laboratories to grow different kinds of microorganisms. A growth or a culture medium is composed of different nutrients that are essential for microbial growth. 3= Inoculum Microbes introduced into a culture medium that initiate growth of organisms. inoculum A small amount of material containing bacteria, viruses, or other microorganisms that is used to start a culture.

What is vertical and horizontal gene transfer and also write type of lateral gene transfer?

1= Vertical Gene Transfer: This is a normal way of transferring genes from parents to offspring. This happens when a cell divide. Each daughter cell receives exactly what its parent cell has. 2= Horizontal Gene Transfer: When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. 3= types of lateral gene transfer: Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include: 1= Transformation, 2= Conjugation, 3= Transduction.

Define Taxonomic Hierarchy:

Taxonomic hierarchy is a systematic arrangement of living organisms into a hierarchical structure based on their evolutionary relationships and shared characteristics. The hierarchy, from broadest to most specific, includes domains, kingdoms, phyla, classes, orders, families, genera, and species. A bacterial species represents —a monophyletic and genomically coherent cluster of individual organisms that show a high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖ (definition taken from the internet). A genus consists of various specie; however, these species differ from each other in certain ways, although these are related by descent with each other. Related genera make up a family. A group of similar families constitutes an order, and a group of similar orders makes up a class. Related classes, in turn, make up a phylum. All phyla that are related to each other make up a kingdom, and related kingdoms are grouped into a domain. Domain ➢ Kingdom ▪ Phylum • Class Order ➢ Family ▪ Genus • Species.

Briefly explain conjugation, transformation, transduction.

1= Conjugation is the process by which one bacterium transfers genetic material to another through direct contact. During conjugation, one bacterium serves as the donor of the genetic material, and the other serves as the recipient Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that replicate independently

of bacterial chromosome. Conjugation requires direct cell to cell contact and cells have to be opposite mating types.

2= Transduction is the process by which foreign DNA is introduced into a bacterial cell by a virus or viral vector. An example is the viral transfer of DNA from one bacterium to another and hence an example of horizontal gene transfer Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released.

3= Bacterial transformation is a process of horizontal gene transfer by which some bacteria take up foreign genetic material (naked DNA) from the environment. ... The prerequisite for bacteria to undergo transformation is its ability to take up free, extracellular genetic material. Such bacteria are termed as competent cells. Naked DNA in solution is transferred from one bacterium to another during this process.

How bacteria can be identified:

A. **Classical or Conventional Method:** Employing culture, microscopy, and biochemical tests to identify bacterial species based on characteristics like morphology and metabolic pathways.

B. **Serological Methods:** Utilizing antibodies and antigens to identify and classify bacteria through specific reactions, such as agglutination or precipitation.

C. **Nucleic Acid-based Methods:** Utilizing DNA or RNA analysis, like polymerase chain reaction (PCR), to identify bacteria based on genetic information, offering precise and rapid identification techniques in modern microbiology.

Describe structural functions following:I. Ribosomes ii. cell wall iii. Flagella.

I. **Ribosomes:** Ribosomes, composed of proteins and ribosomal RNA, facilitate protein synthesis. Prokaryotes possess 70S ribosomes, combining 50S and 30S subunits. Antibiotics like streptomycin target 30S subunits, inhibiting protein synthesis crucial for bacterial growth.

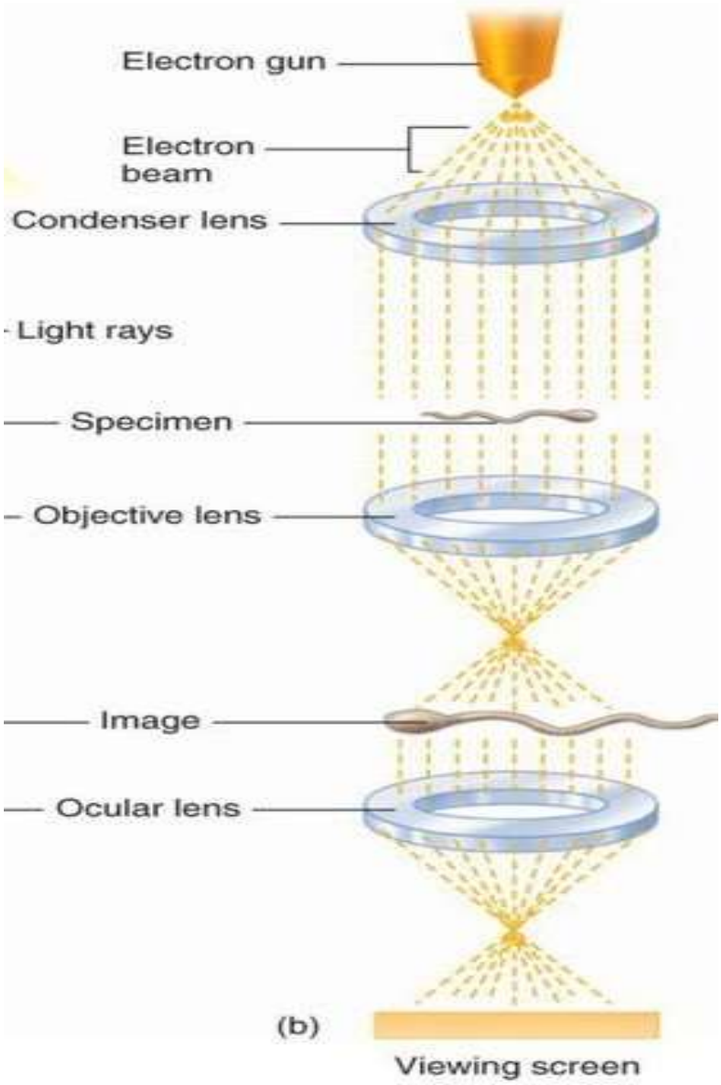
II. **Cell Wall:** The cell wall, a rigid layer outside the plasma membrane, imparts shape, prevents osmotic lysis, and influences pathogenicity in bacteria. It serves as a target for antibiotics, disrupting its integrity and inhibiting bacterial growth.

III. **Flagella:** Flagella, comprising filament, hook, and basal body, propel bacteria. The filament's flagellin protein aids in serovar identification in Gram-negative bacteria.

Types of microbes based on oxygen requirements?

1. **Obligate Aerobes:** These organisms exclusively require oxygen for growth and cannot thrive in its absence. In a nutrient agar tube, their growth is confined to the surface, exemplified by Mycobacterium tuberculosis.
2. **Facultative Anaerobes:** Preferring aerobic respiration, these organisms can switch to anaerobic modes in the absence of oxygen, distributing growth throughout the tube with enhanced development near the surface when oxygen is available.
3. **Obligate Anaerobes:** Unable to use oxygen, these organisms also cannot grow in its presence due to the toxic compounds oxygen forms. Strict anaerobes lack mechanisms to eliminate these toxic compounds.
4. **Aerotolerant Aerobes:** These organisms neither use oxygen nor are affected by its presence, possessing efficient systems to manage and eliminate toxic oxygen compounds.

Label Electron Transmission Microscope:



Define what are the different functions of cell membrane?

The cell membrane serves as a selective permeable barrier, facilitating passive and active transport. In microbes, it orchestrates respiration and photosynthesis. Bactoprenol aids in transporting cell wall components, while the plasma membrane's receptors initiate gene expression signals through ligand binding, regulating various cellular functions.

Define Pilli and also its functions?

Pili are short, hair-like structures on the cell surface of prokaryotic cells. Pili are longer than fimbriae, with one or two per bacterial cell. They serve for attachment to host cells and other bacteria. Pili facilitate DNA transfer during conjugation (Sex pili), contribute to twitching and gliding motility, playing a crucial role in bacterial attachment and motility, distinct from axial filament and fimbriae.

What are the Advantage of Staining:

Staining offers several advantages in microbiology. It enhances microbe visibility, aiding in the observation of shape, size, and arrangements. Staining facilitates the assessment of culture purity or contamination. It enables differentiation and classification of microbes and allows for the detection of various bacterial parts, enhancing microscopic analysis.

What are the Factors that influencing the effectiveness of Antimicrobials:

1= The number of Microbes: It means if there are more numbers of germs in an environment, they will take more time to kill. 2= Environmental Influences: Presence of organic matter such as pus, blood etc. makes antimicrobials less effective. Fats and proteins are especially protective. 3= Time of Exposure: Endospores take more time to get killed. 4= Microbial Characteristics:

Generally, Gram positive bacteria are more sensitive to disinfectants than gram negative microbes. It is because of LPS present in G-Negative bacteria. Mycobacteria are more resistant than other gram-positive microbes. 5= Concentration of Antimicrobials: Concentrated antimicrobial will be more effective than the diluted one.

What are the Subgroup of spiral bacteria:

Spirals are curved shaped bacteria. They are further divided into 3 more subgroups. 1= Vibrio: curved rods, 2= Spirillum: Helical but rigid, 3= Spirochete: Helical but flexible Move by axial filaments.

Write the Name of Three Steps of Glucose Respiration:

Respiration of glucose takes place in three general steps: 1= Glycolysis: Oxidation of glucose to pyruvic acid, 2= Krebs cycle: Oxidation of acetyl CoA to CO2, 3= Electron Transport Chain: Coenzymes that carry electrons from Krebs cycle or glycolysis are oxidized to create ATP.

what are the Characteristics Antibiotic chemical?

Antibiotic chemicals are characterized by their ability to inhibit or kill bacteria, fungi, or other microorganisms. They typically exhibit specificity for certain pathogens, are produced by microorganisms or chemically synthesized, and function by disrupting essential biological processes in the target organisms. One example of chemical antibiotic is penicillin. It is a natural antibiotic produced by the fungus Penicillium and is effective against a wide range of bacteria by inhibiting their cell wall synthesis, leading to cell death.

Five name of bacteria which grow on optimal temperature.

Many bacteria have specific optimal temperature ranges for growth. Here are five examples:

1. Escherichia coli (E. coli): A common bacterium that often resides in the intestines of humans and animals.
2. Staphylococcus aureus: A pathogenic bacterium responsible for various infections.
3. Bacillus subtilis: A soil bacterium with industrial applications.
4. Salmonella enterica: Causes foodborne illnesses in humans.
5. Mycobacterium tuberculosis: The bacterium responsible for tuberculosis in humans.

Define the Medical microbiology?

Medical microbiology is a branch of microbiology focused on human and animal diseases. It involves identifying disease-causing agents, understanding their characteristics, and developing strategies for microbial control and elimination. Additionally, medical microbiologists play a crucial role in tracking down new and emerging pathogens.

What is a passive moment write its types?

A passive moment refers to the movement of substances across a cell membrane without the expenditure of energy by the cell. Types include 1= simple diffusion, 2= facilitated diffusion (aided by transport proteins), and 3= osmosis (movement of water). These processes occur along a concentration gradient, from an area of high concentration to low concentration.

Differences Between Prokaryotes and Eukaryotes.

Three differences between prokaryotes and eukaryotes: 1= Prominent nucleus is found in eukaryotes while it is absent in prokaryotes. 2= Membrane bound organelles are found in eukaryotes and absent in prokaryotes. 3= No histones are found in prokaryotes while present in eukaryotes. 4= Examples of prokaryotes include bacteria and archaea, while all organisms in the domain Eukarya, such as plants and animals, are eukaryotes.

Define Magnetosomes:

Magnetosomes are inclusions of iron oxide surrounded by invaginations of the plasma membrane. Typically found in Gram-negative bacteria, they function as magnets, enabling

bacteria to adhere to iron-containing rocks for nutrition. Additionally, magnetosomes assist in decomposing hydrogen peroxide (H_2O_2), a toxic substance for cells.

Define Oxidative phosphorylation and mechanism of phosphorylation.
Oxidative phosphorylation involves transferring electrons to NAD+ or FAD, initiating a series of electron carriers culminating in oxygen molecules. This electron transport chain generates ATP through chemiosmosis. Phosphorylation, specifically substrate-level phosphorylation, involves directly transferring phosphate to ADP. In photosynthetic cells, photophosphorylation occurs, where light energy is converted into ATP through a process analogous to oxidative phosphorylation.

Define Catalase and its protocol:

Catalase is an enzyme that catalyzes the decomposition of hydrogen peroxide into water and oxygen. In a typical protocol, a sample is mixed with hydrogen peroxide, and the rate of oxygen release is measured. Increased oxygen production indicates higher catalase activity, providing insights into cellular antioxidant defense mechanisms and metabolic processes.